

0.1 Exact functors

Let $F : \mathcal{A} \rightarrow \mathcal{A}'$ be an additive functor.

Definition 0.1. F is called

- (a) *left exact* $\iff F$ commutes with finite limits.
- (b) *right exact* $\iff F$ commutes with finite colimits.
- (c) *exact* $\iff F$ is left and right exact.

Remark. Since $\mathcal{A}, \mathcal{A}'$ are additive categories, all finite limits and colimits exist in $\mathcal{A}$ and $\mathcal{A}'$. So if $D : J \rightarrow \mathcal{A}$ is a finite diagram, we have $\lim_J D$ exists in $\mathcal{A}$, $\lim_J F \circ D$ exists in $\mathcal{A}'$ and we have a natural morphism

$$F(\lim_J D) \rightarrow \lim_J F \circ D$$

in $\mathcal{A}'$. F is left exact if this morphism is an isomorphism $\forall D$.

Proposition 0.2. Let $F : \mathcal{A} \rightarrow \mathcal{A}'$ be additive, then:

(a) The following are equivalent:

- (i) F is left exact.
- (ii) F commutes with the formation of kernels, i.e. $\forall f : X \rightarrow Y$ in $\mathcal{A}$, the natural morphism

$$F(\ker f) \rightarrow \ker F(f)$$

is an isomorphism.

- (iii) $\forall$ exact sequence $0 \rightarrow X' \rightarrow X \rightarrow X''$ in $\mathcal{A}$, the sequence $0 \rightarrow FX' \rightarrow FX \rightarrow FX''$ is exact in $\mathcal{A}'$.
- (iv) $\forall$ exact sequence $0 \rightarrow X' \rightarrow X \rightarrow X'' \rightarrow 0$ in $\mathcal{A}$, the sequence $0 \rightarrow FX' \rightarrow FX \rightarrow FX''$ is exact in $\mathcal{A}'$.

(b) The following are equivalent:

- (i) F is right exact.
- (ii) F commutes with the formation of cokernels.
- (iii) $\forall$ exact sequence $X' \rightarrow X \rightarrow X'' \rightarrow 0$ in $\mathcal{A}$, the sequence $FX' \rightarrow FX \rightarrow FX'' \rightarrow 0$ is exact in $\mathcal{A}'$.
- (iv) $\forall$ exact sequence $0 \rightarrow X' \rightarrow X \rightarrow X'' \rightarrow 0$ in $\mathcal{A}$, the sequence $FX' \rightarrow FX \rightarrow FX'' \rightarrow 0$ is exact in $\mathcal{A}'$.

(c) F is exact $\iff \forall$ exact sequences $X' \xrightarrow{f} X \xrightarrow{g} X''$, the sequence

$$FX' \xrightarrow{Ff} FX \xrightarrow{Fg} FX''$$

is exact.

Proof. TODO. □

Proposition 0.3. $\forall X \in \mathcal{A}$ the co- and contravariant Hom functors $\text{Hom}_{\mathcal{A}}(X, -), \text{Hom}_{\mathcal{A}}(-, X)$ are left exact.

Proof. TODO. □

Proposition 0.4. *Let $F : \mathcal{A} \rightarrow \mathcal{A}'$ and $G : \mathcal{A}' \rightarrow \mathcal{A}$ be additive functors with $F \dashv G$. Then F is right exact and G is left exact.*

Proof. TODO. □

Example 0.5. By Hom- $\otimes$ Adjunction (roughly $\text{Hom}(M \otimes -, N) = \text{Hom}(M, \text{Hom}(-, N))$) $M \otimes -$ is left adjoint to $\text{Hom}(-,) \implies M \otimes_R - : {}_R\text{Mod} \rightarrow \mathbf{Ab}$ is right exact, as is $- \otimes_R M : \text{Mod}_R \rightarrow \mathbf{Ab}$.

$$(- \otimes_R M : \text{Mod}_R \rightarrow \mathbf{Ab}) = (M \otimes_{R^{\text{op}}} - : {}_{R^{\text{op}}}\text{Mod} \rightarrow \mathbf{Ab})$$

Remark. Next small goal: J any small index category. Are $\lim_J : \mathcal{A}^J \rightarrow \mathcal{A}$, $\text{colim}_J : \mathcal{A}^J \rightarrow \mathcal{A}$ exact?

Proposition 0.6. $\mathcal{A}^J = \text{Fun}(J, \mathcal{A})$ is an abelian category.

Proposition 0.7. Suppose $\mathcal{A}$ contains all limits (colimits) for a given small index category J , then $\lim_J : \mathcal{A}^J \rightarrow \mathcal{A}$ is left exact. ($\text{colim}_J : \mathcal{A}^J \rightarrow \mathcal{A}$ is right exact.)

Corollary 0.8. Let I be a set and $\underline{I}$ the discrete category associated to I . Then:

- (a) $\prod_{i \in I} : A^I \rightarrow A, (A_i)_{i \in I} \mapsto \prod_{i \in I} A_i$
- (b) $\bigoplus_{i \in I} : A^I \rightarrow A, (A_i)_{i \in I} \mapsto \bigoplus_{i \in I} A_i$ assuming existence, are exact functors.

Definition 0.9. (a) A non-empty category J is called *filtered* iff

$$(i) \forall i, j \in J : \exists \text{ diagram } \begin{array}{ccc} & i & \\ & \searrow & \\ j & \longrightarrow & k \end{array} \in J.$$

(ii) $\forall f, g : i \rightrightarrows j \exists h : j \rightarrow k$ such that

$$h \circ f = h \circ g : i \rightarrow k$$

- (b) A directed poset $(I, \subseteq)$ is a poset such that $\forall i, j \in I \exists h \in I$ such that $h \geq i, h \geq j$ ($\implies (I, \subseteq)$ is directed $\implies \text{ord}(I, \subseteq)$ is filtered.)

Definition 0.10. $\mathcal{A}$ has exact filtered colimits $\iff$ for each filtered index category J , the functor $\text{colim } \mathcal{A}^J \rightarrow \mathcal{A}$ is exact (and defined).

Theorem 0.11. ${}_R\text{Mod}$ has exact filtered colimits.

Fundamental question: how to investigate the non-exactness of functors (that are left or right exact). Answer of homological algebra: e.g. $F : A \rightarrow B$ left exact, define “higher right derived functors” $(R^i F)_{i \geq 0}$ from F such that $\forall$ s.e.s.

$$0 \rightarrow X' \rightarrow X \rightarrow X'' \rightarrow 0$$

in $\mathcal{A}$

$$\begin{array}{ccccccc}
0 & \longrightarrow & FX' & \longrightarrow & FX & \longrightarrow & FX'' \\
& & & & & \searrow & \\
& & R^1FX' & \longrightarrow & R^1FX & \longrightarrow & R^1FX'' \\
& & & & & \searrow & \\
& & R^2FX' & \longrightarrow & R^2FX & \longrightarrow & R^2FX'' \longrightarrow \dots
\end{array}$$

and $R^0F = F$. Study R^iF to understand the nonexactness of F , or to gain insight into some invariants of $\mathcal{A}$. Some R^iF (typically $i \leq 3$) have concrete meanings.

0.1.1 To define R^iF (or L_iF)

One wants “enough” injectives (projectives) in $\mathcal{A}$.

Theorem-Definition 0.12. *For $I \in \mathcal{A}$ the following are equivalent:*

- (i) $\forall$ diagrams with ι monomorphism $\exists$ extension $g : B \rightarrow I$ such that the following commutes

$$\begin{array}{ccccc}
0 & \longrightarrow & A & \xrightarrow{\iota} & B \\
& & \downarrow f & \swarrow g & \\
& & I & &
\end{array}$$

- (ii) The functor $\text{Hom}_{\mathcal{A}}(-, I) : \mathcal{A}^{\text{op}} \rightarrow \mathbf{Ab}$ is exact.

- (iii) Every s.e.s. $0 \rightarrow I \xrightarrow{h} C \xrightarrow{k} D \rightarrow 0$ in $\mathcal{A}$ is split.

If any of these hold then I is called an injective object.

Theorem-Definition 0.13. *For $P \in \mathcal{A}$ the following are equivalent:*

- (i) $\forall$ diagrams with π epimorphism $\exists$ lifting $g : P \rightarrow B$ such that $\pi \circ g = f$

$$\begin{array}{ccccc}
& & P & & \\
& \swarrow g & \downarrow f & & \\
B & \xrightarrow{\pi} & C & \longrightarrow & 0
\end{array}$$

- (ii) The functor $\text{Hom}_{\mathcal{A}}(P, -) : \mathcal{A} \rightarrow \mathbf{Ab}$ is exact.

- (iii) Every s.e.s. $0 \rightarrow A \rightarrow B \rightarrow P \rightarrow 0$ is split.

If these hold, then P is called a projective object in $\mathcal{A}$.

Remark. I injective in $\mathcal{A} \iff I$ projective in $\mathcal{A}^{\text{op}}$.

Proposition 0.14. (a) If $(I_j)_{j \in J}$ (J a set) is a family of injectives in $\mathcal{A}$, such that $\prod_j I_j$ exist, then $\prod_j I_j$ is injective.

(b) If $(P_j)_{j \in J}$ (J a set) is a family of projectives in $\mathcal{A}$, such that $\bigoplus_{j \in J} P_j$ exist, then $\bigoplus_j P_j$ is projective.

Example 0.15. (a) R is a projective R -module (hence so is $R^{(I)}$)

(b) $\mathbb{Q}/\mathbb{Z}$ is an injective in ${}_{\mathbb{Z}}\mathbf{Mod}$.

Definition 0.16. (a) $\mathcal{A}$ has enough injective $\iff \forall X \in \mathcal{A} \exists$ monomorphism $X \rightarrow I$ with $I \in \mathcal{A}$ injective.

(b) $\mathcal{A}$ has enough projectives $\iff \forall X \in \mathcal{A} \exists$ epimorphism $P \rightarrow X$ with $P \in \mathcal{A}$ projective.

Definition 0.17. $Q \in \mathcal{A}$ is called a

(a) *generator* $\iff \text{Hom}_{\mathcal{A}}(Q, -) : \mathcal{A} \rightarrow \mathbf{Ab}$ is faithful.

(b) *cogenerator* $\iff \text{Hom}_{\mathcal{A}}(-, Q) : \mathcal{A}^{\text{op}} \rightarrow \mathbf{Ab}$ is faithful.

Remark 0.18. For $Q \in \mathcal{A}$ the following are equivalent:

1. Q is a generator.
2. $\forall X, Y \in \mathcal{A} \forall f, g \in \mathcal{A}(X, Y) :$

$$f \neq g \implies \exists h \in \mathcal{A}(Q, X) : f \circ h \neq g \circ h$$

3. $\forall X, Y \in \mathcal{A} \forall f \in \mathcal{A}(X, Y) :$

$$f \neq 0 \implies \exists h \in \mathcal{A}(Q, X) : f \circ h \neq 0$$

Examples 0.19. 1. R is a generator for ${}_R\mathbf{Mod}$: Suppose $f : M \rightarrow N$ in ${}_R\mathbf{Mod}$ is nonzero. Then $\exists m \in M : f(m) \neq 0$. Define $h : R \rightarrow M, r \mapsto rm \implies f \circ h : R \rightarrow N$ is nonzero.

2. $\mathbb{Q}/\mathbb{Z}$ is a cogenerator in $\mathbf{Ab} = {}_{\mathbb{Z}}\mathbf{Mod}$ let $f : M \rightarrow N$ be non-zero in $\mathbf{Ab}$. Let $X \in \text{im}(f) \setminus \{0\}$. TODO

Definition 0.20. $\mathcal{A}$ is called a *Grothendieck abelian category* (GAC) iff

1. $\mathcal{A}$ is cocomplete.
2. For any filtered small category $J : \text{colim} : A^J \rightarrow \mathcal{A}$ is exact.
3. $\mathcal{A}$ possesses a cogenerator.

Theorem 0.21. For a GAC $\mathcal{A}$ the following hold:

1. The subobjects and quotient objects of any $A \in \mathcal{A}$ form a set.
2. $\mathcal{A}$ has enough injectives.
3. $\mathcal{A}$ has an injective cogenerator.
4. $\mathcal{A}$ is complete.

Example 0.22. 1. ${}_R\mathbf{Mod}$ is a GAC by examples 19(a) Thm 11 Cor II48

2. If $\mathcal{A}$ is a GAC and J is a small category then $\mathbf{PSh}(J, \mathcal{A})$ is a GAC

Lemma 0.23. Suppose $F : \mathcal{A} \rightarrow \mathcal{A}'$ and $G : \mathcal{A}' \rightarrow \mathcal{A}$ are functors such that $F \dashv G$, then:

1. F exact $\implies G$ maps injectives in $\mathcal{A}'$ to injectives in $\mathcal{A}$.
2. G exact $\implies F$ maps projectives in $\mathcal{A}$ to projectives in $\mathcal{A}'$.
3. F faithful $\implies G$ maps cogenerators in $\mathcal{A}'$ to cogenerators in $\mathcal{A}$.
4. G faithful $\implies F$ maps generators in $\mathcal{A}$ to generators in $\mathcal{A}'$.

Theorem 0.24. Let $\mathcal{A}$ be a GAC with generator Q , then:

Corollary 0.25. $\mathbb{Q}/\mathbb{Z}$ is injective in $\mathbf{Ab}$.

Corollary 0.26. For any ring R , $\text{Hom}(R, \mathbb{Q}/\mathbb{Z})$ is an injective R -module and a cogenerator.

Proposition 0.27. Let $Q, J \in \mathcal{A}$, then

1. Suppose $\mathcal{A}$ contains all coproducts over index sets. Then the following are equivalent:
 - (a) Q is a generator.
 - (b) $\forall X \in \mathcal{A} \exists$ set $I, \exists$ epimorphism $Q^{(I)} := \coprod_{i \in I} Q \rightarrow X$. Moreover, if Q is a rojective generator then $\mathcal{A}$ has enough projectievs of the form $Q^{(I)}$, where I is a set.
2. Suppose $\mathcal{A}$ contains all products over index sets then the following are equivalent:
 - (a) J is a cogenerator.
 - (b) $\forall Y \in \mathcal{A} \exists$ set I , monomorphism $Y \rightarrow J^I := \prod_{i \in I} J$. Moreover if J is an injective cogenerrator then $\mathcal{A}$ has enough injectives of the form J^I where I is a set.

Corollary 0.28. ${}_R\mathbf{Mod}$ has enough injectives of the form $\text{Hom}_{\mathbb{Z}}(R, \mathbb{Q}/\mathbb{Z})^I$, I a set.

Remark 0.29. TODO

0.2 (Co-)chain complexes, (co-)homology

Let $\mathcal{A}$ be an additive category.

Definition 0.30. (a) A chain complex (C_*, ∂_*) over $\mathcal{A}$ is a sequence $(\partial_i : C_i \rightarrow C_{i-1})_{i \in \mathbb{Z}}$ of morphisms in $\mathcal{A}$ such that $\partial_i \circ \partial_{i+1} = 0, \forall i \in \mathbb{Z}$. The map ∂_i is called the i -th differential or i -th boundary map of the complex. A morphism $f_* : C_* \rightarrow D_*$ of chain complexes is a sequence of morphisms $f_* = (f_i : C_i \rightarrow D_i)_{i \in \mathbb{Z}}$ such that the diagram commutes $\forall i \in \mathbb{Z}$

$$\begin{array}{ccc} C_{i-1} & \xleftarrow{\partial_i^C} & C_i \\ f_{i-1} \downarrow & & \downarrow f_i \\ D_{i-1} & \xleftarrow{\partial_i^D} & D_i \end{array}$$

The category of chain complexes over $\mathcal{A}$ is denoted $\text{Ch}_*(\mathcal{A})$.

- (b) A *cochain complex* (C^*, δ^*) over $\mathcal{A}$ is a sequence $(\delta^i : C^i \rightarrow C^{i+1})_{i \in \mathbb{Z}}$ of morphisms in $\mathcal{A}$, such that $\delta^{i+1} \circ \delta^i = 0, \forall i \in \mathbb{Z}$. A morphism $f^* : C^* \rightarrow D^*$ of cochain complexes is a sequence $f^* = (f^i : C^i \rightarrow D^i)_{i \in \mathbb{Z}}$ of morphisms in $\mathcal{A}$ such that the following diagram commutes $\forall i \in \mathbb{Z}$

$$\begin{array}{ccc} D_i & \xrightarrow{\delta_D^i} & D_{i+1} \\ f_i \uparrow & & \uparrow f_{i+1} \\ C_i & \xrightarrow{\delta_C^i} & C_{i+1} \end{array}$$

The *category of cochain complexes* over $\mathcal{A}$ is denoted $\text{Ch}^*(\mathcal{A})$

Exercise 0.31. TODO

In the following we do most things only for cochain complexes.

- Definition 0.32.** (a) The *support* of a cochain complex $C^* \in \text{Ch}^*(\mathcal{A})$ is $\text{supp } C = \{i \in \mathbb{Z} \mid C^i \neq 0\}$
- (b) The full subcategory of $\text{Ch}^*(\mathcal{A})$ on complexes supported on $\mathbb{N}_0$ (or on $-\mathbb{N}_0$) is denoted $\text{Ch}_{\geq 0}^*(\mathcal{A})$ (or $\text{Ch}_{\leq 0}^*(\mathcal{A})$).

Proposition 0.33. If $\mathcal{A}$ is additive (or abelian), then so are $\text{Ch}^*(\mathcal{A}), \text{Ch}_{\geq 0}^*(\mathcal{A}), \text{Ch}_{\leq 0}^*(\mathcal{A})$.

Definition 0.34. For $i \in \mathbb{Z}$ define left shift by i by

$$\begin{array}{ccc} \text{Ch}^*(\mathcal{A}) & \longrightarrow & \text{Ch}^*(\mathcal{A}) \\ C & \longmapsto & C[i] \\ f \downarrow & & \downarrow f[i] \\ D & \longmapsto & D[i] \end{array}$$

where $C[i]^n = C^{n+i}, \delta_{C[i]}^n = \delta_C^{n+i}$ and $f[i]^n = f^{n+1}$.

Convention 0.35. We regard $\mathcal{A}$ as a subcategory of $\text{Ch}^*(\mathcal{A})$, as the subcategory of complexes C^* with $\text{supp } C^* \subseteq \{0\}$. Identify $X \in \mathcal{A}$ with complex

$$\cdots \rightarrow 0 \rightarrow 0 \rightarrow X \rightarrow 0 \rightarrow 0 \rightarrow \cdots$$

Example 0.36. TODO

0.3 Double complexes

Can iterate the formation of $\mathcal{A} \rightarrow \text{Ch}^*(\mathcal{A})$.

Definition 0.37. The category of double (cochain) complexes is $\text{Ch}^{**}(\mathcal{A}) := \text{Ch}^*(\text{Ch}^*(\mathcal{A}))$, so objects of $\text{Ch}^{**}(\mathcal{A})$ are complexes of complexes

Definition 0.38. $C \in \text{Ch}^{**}(\mathcal{A})$ is called *bounded* $\iff \forall k \in \mathbb{Z} : \#\{(i, j) \in \mathbb{Z}^2 \mid i + j = k, C^{ij} \neq 0\} < \infty$. Write $\text{Ch}_b^{**}(\mathcal{A}) \subseteq \text{Ch}^{**}(\mathcal{A})$ for the full subcategory on bounded double complexes.

Exercise 0.39. If $\mathcal{A}$ is additive or abelian, then so is $\text{Ch}_b^{**}(\mathcal{A})$.

Definition 0.40. The *total complex* $\text{Tot}(C)$ of $C \in \text{Ch}_b^{**}(\mathcal{A})$ is the complex $\overline{C} \in \text{Ch}^*(\mathcal{A})$ defined as follows:

$$\overline{C}^k := \bigoplus_{\substack{(i,j) \in \mathbb{Z}^2 \\ i+j=k}} C^{ij}$$

$\delta_{\overline{C}}^k$ is constructed as follows:

$$\delta^{ij} : C^{ij} \xrightarrow{(\delta_1^{ij}, (-1)^i \delta_2^{ij})} C^{i+1,j} \oplus C^{i,j+1} \hookrightarrow \bigoplus_{i'+j'=k+1} C^{i'j'} = \overline{C}^{k+1}$$

Use the universal property of the direct sum to define

$$\delta_{\overline{C}}^k = \bigoplus_{i+j=k} \delta^{ij} : \overline{C}^k \rightarrow \overline{C}^{k+1}$$

TODO

Exercise 0.41. TODO

Definition 0.42. For $C = (C^*, \delta^*) \in \text{Ch}^*(\mathcal{A})$ define:

- $Z^i(C) := \ker(\delta^i)$ as the i -th cocycle object.
- $B^i(C) := \text{im}(\delta^{i-1})$ as the i -th coboundary object.
- $u^i(C) : B^i(C) \rightarrow Z^i(C)$ the canonical monomorphism.
- $H^i(C) := \text{coker}(u^i(C)) = Z^i(C)/B^i(C)$ as the i -th cohomology object.

(Co-)homology measures the non-exactness of the complex C .

Lemma 0.43 (Alternative description of cohomology object). *For $C \in \text{Ch}^*(\mathcal{A})$ consider TODO*

Lemma 0.44 (exer). *TODO*

Theorem 0.45. (a) *Given a s.e.s.*

$$\mathcal{E} : 0 \rightarrow C \xrightarrow{f} D \xrightarrow{g} E \rightarrow 0$$

one obtains a long exact sequence

$$\dots \rightarrow H^i(C) \xrightarrow{H^i(f)} H^i(D) \xrightarrow{H^i(g)} H^i(E) \xrightarrow{d_{\mathcal{E}}^i} H^{i+1}(C) \xrightarrow{H^{i+1}(f)} \dots$$

for $i \in \mathbb{Z}$, where the connecting homomorphism $d_{\mathcal{E}}^i$ is defined by the snake lemma and $\mathcal{E} \mapsto d_{\mathcal{E}}^i$ is “functorial”.

(b) *Given a morphism of short exact sequences $\varphi : \mathcal{E} \rightarrow \mathcal{E}'$ in $\text{Ch}^*(\mathcal{A})$:*

$$\begin{array}{ccccccccc} \mathcal{E} : 0 & \longrightarrow & C & \xrightarrow{f} & D & \xrightarrow{g} & E & \longrightarrow & 0 \\ \varphi \downarrow & & \downarrow \alpha & & \downarrow \beta & & \downarrow \gamma & & \\ \mathcal{E}' : 0 & \longrightarrow & C' & \xrightarrow{f'} & D' & \xrightarrow{g'} & E' & \longrightarrow & 0 \end{array}$$

one obtains a commutative ladder of long exact sequences from (a)

$$\begin{array}{ccccccc}
\cdots & \longrightarrow & H^i(C) & \xrightarrow{H^i(f)} & H^i(D) & \xrightarrow{H^i(g)} & H^i(E) & \xrightarrow{d_{\mathcal{E}}^i} & H^{i+1}(C) & \longrightarrow & \cdots \\
& & \downarrow H^i(\alpha) & & \downarrow H^i(\beta) & & \downarrow H^i(\gamma) & & \downarrow H^{i+1}(\alpha) & & \\
\cdots & \longrightarrow & H^i(C') & \xrightarrow{H^i(f')} & H^i(D') & \xrightarrow{H^i(g')} & H^i(E') & \xrightarrow{d_{\mathcal{E}'}^i} & H^{i+1}(C') & \longrightarrow & \cdots
\end{array}$$

Definition 0.46. $C \in \text{Ch}^*(\mathcal{A})$ is called *acyclic* if C is exact $\iff H^i(C) = 0, \forall i \in \mathbb{Z}$.

Corollary 0.47 (To theorem 45). *Let $\mathcal{E} : 0 \rightarrow C' \rightarrow C \rightarrow C'' \rightarrow 0$ be a s.e.s. in $\text{Ch}^*(\mathcal{A})$, then if any two of C', C, C'' are acyclic, then so is the third.*

Theorem 0.48 (Acyclicity criterion for total complexes). *Let $C \in \text{Ch}_b^{**}(\mathcal{A})$ such that*

(a) *Each row $(C^{ij}, \delta_1^{ij})_{i \in \mathbb{Z}}$ is acyclic $\forall j \in \mathbb{Z}$, or*

(b) *Each column $(C^{ij}, \delta_1^{ij})_{j \in \mathbb{Z}}$ is acyclic $\forall i \in \mathbb{Z}$*

then $\text{Tot}(C)$ is acyclic.

Definition 0.49. An arrow $f : C \rightarrow D$ in $\text{Ch}^*(\mathcal{A})$ is called a *quasi-isomorphism* (qism) $\iff \forall i \in \mathbb{Z} : H^i(f) : H^i(C) \rightarrow H^i(D)$ is an isomorphism.

Lemma 0.50. *TODO*

Corollary 0.51. *For $f : C \rightarrow D$ in $\text{Ch}^*(\mathcal{A})$ the following are equivalent:*

(a) *f is a quasi-isomorphism.*

(b) *Cone f is acyclic.*

Definition 0.52. Let $f, g : C \rightarrow D$ in $\text{Ch}^*(\mathcal{A})$.

(a) A *homotopy* from f to g is a sequence of morphisms $(s^i : C^i \rightarrow D^{i-1})_{i \in \mathbb{Z}}$ such that $\forall i \in \mathbb{Z} : f^i - g^i = \delta_D^{i-1} \circ s^i + s^{i+1} \circ \delta_C^i$, i.e.

$$\begin{array}{ccccc}
& & C^i & \xrightarrow{\delta_C^i} & C^{i+1} \\
& \swarrow s^i & \downarrow f^i - g^i & \swarrow s^{i+1} & \\
D^{i-1} & \xrightarrow{\delta_D^{i-1}} & D^i & &
\end{array}$$

(b) f is called *homotopic* to g if $\exists$ homotopy from f to g . Write $f \sim g$.

(c) f is called *nullhomotopic* if $f \sim 0$.

Note: This definition only requires that $\mathcal{A}$ is additive.

Proposition 0.53. (a) *For $C, D \in \text{Ch}^*(\mathcal{A})$, homotopy defines an equivalence relation on $\text{Hom}_{\text{Ch}^*(\mathcal{A})}(C, D)$.*

(b) *For $f, f' : C \rightarrow D$ and $g, g' : D \rightarrow E$ in $\text{Ch}^* \mathcal{A}$ one has:*

$$f \sim f', g \sim g' \implies g \circ f \sim g' \circ f'$$

(c) Suppose $F : \mathcal{A} \rightarrow \mathcal{B}$ is an additive functor, then one has a functor $F : \text{Ch}^* \mathcal{A} \rightarrow \text{Ch}^* \mathcal{B}$ by

$$F(\delta^n : C^n \rightarrow C^{n+1})_{n \in \mathbb{Z}} = (F\delta^n : FC^n \rightarrow FC^{n+1})_{n \in \mathbb{Z}}$$

This functor preserves homotopy, i.e. $f \sim g \implies Ff \sim Fg$.

Proposition 0.54. Suppose $f, g : C \rightarrow D$ in $\text{Ch}^* \mathcal{A}$ are homotopic, then

$$H^i(f) = H^i(g) : H^i(C) \rightarrow H^i(D), \forall i \in \mathbb{Z}$$

Definition 0.55. A morphism $f : C \rightarrow D$ in $\text{Ch}^* \mathcal{A}$ is called a *homotopy equivalence* from C to D if $\exists g : D \rightarrow C$ in $\text{Ch}^* \mathcal{A}$ such that

$$g \circ f \sim 1_C, \quad f \circ g \sim 1_D$$

and in this case C and D are called homotopy equivalent.

Proposition 0.56. Suppose $f : C \rightarrow D$ is a homotopy equivalence, then f is a quasi-isomorphism.

Example. TODO

0.4 Injective and projective resolutions

Notation. Inj or $\text{Inj}_{\mathcal{A}}$ and Proj or $\text{Proj}_{\mathcal{A}}$ are the full subcategories of $\mathcal{A}$ on injective or projective objects respectively. Note that $\text{Inj}_{\mathcal{A}}^{\text{op}} = \text{Proj}_{\mathcal{A}^{\text{op}}}$.

Definition 0.57. (a) An *injective resolution* of $A \in \mathcal{A}$ is a quism $f : \underline{A} \rightarrow I$ in $\text{Ch}_{\geq 0}^* \mathcal{A}$ with $I \in \text{Ch}_{\geq 0}^*(\text{Inj}_{\mathcal{A}})$ i.e. $\text{Cone } f$, which is

$$0 \rightarrow A \rightarrow I^0 \rightarrow I^1 \rightarrow I^2 \rightarrow \dots$$

is acyclic and all the I^j are injective.

(b) A *projective resolution* of $A \in \mathcal{A}$ is a quism $g : P \rightarrow \underline{A}$ in $\text{Ch}_{\leq 0}^* \mathcal{A}$ with $P \in \text{Ch}_{\leq 0}^*(\text{Proj}_{\mathcal{A}})$.

Proposition 0.58. Consider the functor

$$\begin{array}{ccc} \hat{\cdot} : \text{Ch}^*(\mathcal{A})^{\text{op}} & \longrightarrow & \text{Ch}^*(\mathcal{A}^{\text{op}}) \\ (C^n, \delta_C^n) & \xrightarrow{\hat{\cdot}} & (\hat{C}^n, \hat{\delta}_C^n) \\ f \downarrow & & \uparrow \hat{f} \\ (D^n, \delta_D^n) & \xrightarrow{\hat{\cdot}} & (\hat{D}^n, \hat{\delta}_D^n) \end{array}$$

where $\hat{C}^n = C^{-n}$, $\hat{\delta}_C^n = \delta^{-n-1} \in \mathcal{A}(C^{-n-1}, C^{-n}) = \mathcal{A}^{\text{op}}(\hat{C}^n, \hat{C}^{n+1})$ and $\hat{f}^n := f^{-n} \in \mathcal{A}(C^{-n}, D^{-n}) = \mathcal{A}^{\text{op}}(\hat{D}^n, \hat{C}^n)$. Then

(a) $\hat{\cdot}$ is well defined, satisfies $\hat{\cdot} \circ \hat{\cdot} = \text{id}$ and $\hat{\cdot}$ is an isomorphism of categories.

(b) $\hat{\cdot}(\text{Ch}_{\geq 0/\leq 0}^*(\mathcal{A})^{\text{op}}) = \text{Ch}_{\leq 0/\geq 0}^*(\mathcal{A}^{\text{op}})$.

$$(c) \sim (\text{Ch}_{\geq 0/\leq 0}^*(\text{Inj}_{\mathcal{A}} / \text{Proj}_{\mathcal{A}}))^{\text{op}} = \text{Ch}_{\leq 0/\geq 0}^*(\text{Proj}_{\mathcal{A}^{\text{op}}} / \text{Inj}_{\mathcal{A}^{\text{op}}})$$

(d) If $\underline{A} \rightarrow I$ is an injective resolution in $\text{Ch}^* \geq 0(\mathcal{A})$, then $\widehat{I} \rightarrow \underline{A}$ is a projective resolution in $\text{Ch}_{\leq 0}^*(\mathcal{A}^{\text{op}})$

Theorem 0.59. Let $\mathcal{A}$ be an abelian category with enough injectives, then

(a) Each $A \in \mathcal{A}$ possesses an injective resolution.

(b) Let $h : A \rightarrow B$ be a morphism, let $f : \underline{B} \rightarrow I^\bullet$ be an injective resolution and $g : \underline{A} \rightarrow C^\bullet$ be a quism in $\text{Ch}_{\geq 0}^*(\mathcal{A})$ ($C^\bullet$ is a resolution of A), then there is a commutative diagram in $\text{Ch}_{\geq 0}^*\mathcal{A}$

$$\begin{array}{ccc} \underline{A} & \xrightarrow{g} & C^\bullet \\ h \downarrow & & \downarrow H \\ \underline{B} & \xrightarrow{f} & I^\bullet \end{array}$$

(c) If diagram in (b) commutes with $H, H' : C^\bullet \rightarrow I^\bullet$, then $H' \sim H$.

Corollary 0.60. Suppose $\underline{A} \xrightarrow{g} I^\bullet$ and $\underline{A} \xrightarrow{g} J^\bullet$ are inj resolutions of A . Then:

(a) $\exists H : I^\bullet \rightarrow J^\bullet$ such that

$$\begin{array}{ccc} & \underline{A} & \\ g \swarrow & & \searrow f \\ J^\bullet & \xleftarrow{\quad H \quad} & I^\bullet \end{array}$$

commutes.

(b) H in (a) is always a homotopy equivalence.

Lemma 0.61 (Horseshoe 2). Let $0 \rightarrow A' \rightarrow A \rightarrow A'' \rightarrow 0$ be a s.e.s. in $\mathcal{A}$, let $\underline{A'} \xrightarrow{f'} I^\bullet$ and $\underline{A''} \xrightarrow{f''} J^\bullet$ be injective resolutions. Then $\exists$ commutative diagram:

$$\begin{array}{ccccccc} 0 & \longrightarrow & \underline{A'} & \longrightarrow & \underline{A} & \longrightarrow & \underline{A''} \longrightarrow 0 \\ & & \downarrow f' & & \downarrow f & & \downarrow f'' \\ 0 & \longrightarrow & I^\bullet & \longrightarrow & K^\bullet & \longrightarrow & J^\bullet \longrightarrow 0 \end{array}$$

in $\text{Ch}_{\geq 0}^*\mathcal{A}$ with exact rows and an injective resolution $f : \underline{A} \rightarrow K^\bullet$. Moreover $\forall n \geq 0$ the following commutative diagram has exact rows and columns:

$$\begin{array}{ccccccc} & & 0 & & 0 & & 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & \text{im}(\delta_I^{n-1}) & \longrightarrow & \text{im}(\delta_K^{n-1}) & \longrightarrow & \text{im}(\delta_J^{n-1}) \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & I^n & \longrightarrow & K^n & \longrightarrow & J^n \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & \text{im}(\delta_I^n) & \longrightarrow & \text{im}(\delta_K^n) & \longrightarrow & \text{im}(\delta_J^n) \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ & & 0 & & 0 & & 0 \end{array}$$

So $f' = \delta_I^{-1}$, $f = \delta_K^{-1}$ and $f'' = \delta_J^{-1}$.

Definition 0.62. Define $\text{Ex}_{\mathcal{A}}$ as the category of s.e.s. in $\mathcal{A}$ with objects:

$$\mathcal{E}_{\mathcal{A}} : 0 \rightarrow A' \rightarrow A \rightarrow A'' \rightarrow 0$$

in $\mathcal{A}$, and morphisms are commutative diagrams

$$\begin{array}{ccccccccc} 0 & \longrightarrow & A' & \longrightarrow & A & \longrightarrow & A'' & \longrightarrow & 0 \\ & & \downarrow f' & & \downarrow f & & \downarrow f'' & & \\ 0 & \longrightarrow & B' & \longrightarrow & B & \longrightarrow & B'' & \longrightarrow & 0 \end{array}$$

in $\mathcal{A}$ with exact rows represented by $\underline{f} = (f', f, f'')$ in $\text{Ex}_{\mathcal{A}}(\mathcal{E}_A, \mathcal{E}_B)$. Composition of arrows is componentwise. TODO

Proposition 0.63. (a) $\text{Ex}_{\mathcal{A}}$ is an additive category.

(b) We have additive functors $\text{pr}_i : \text{Ex}_{\mathcal{A}} \rightarrow \mathcal{A}$, mapping $0 \rightarrow A_1 \rightarrow A_2 \rightarrow A_3 \rightarrow 0$ to A_i .

(c) $\text{Ex}_{\mathcal{A}}$ is not abelian.

Definition 0.64. An arrow $\underline{f} = (f', f, f'')$ in $\text{Ex}_{\mathcal{A}}$ is called

(a) a *strict monomorphism (strict epimorphism)* $\iff f', f, f''$ are monics (epics) in $\mathcal{A}$.

(b) *strict* $\iff$

$$0 \rightarrow \ker f' \rightarrow \ker f \rightarrow \ker f'' \rightarrow 0$$

is exact which is (by the snake lemma) equivalent to exactness of

$$0 \rightarrow \text{coker } f' \rightarrow \text{coker } f \rightarrow \text{coker } f'' \rightarrow 0$$

Proposition 0.65. If $\underline{f}$ is strict in $\text{Mor}(\text{Ex}_{\mathcal{A}})$ then $\ker \underline{f}$, $\text{coker } \underline{f}$, $\text{im } \underline{f}$, $\text{coim } \underline{f}$ exist in $\text{Ex}_{\mathcal{A}}$ and the canonical map $\text{coim } \underline{f} \rightarrow \text{im } \underline{f}$ is an isomorphism.

Remark. $\text{Ex}_{\mathcal{A}}$ is an exact category.

Definition 0.66. (a) A complex $\mathcal{E}^\bullet = (\underline{\delta}^i :)$ TODO

Theorem 0.67. TODO

0.5 Derived Functors

Let $\mathcal{A}, \mathcal{B}$ be abelian categories.

Definition 0.68. (a) A homological (resp. cohomological) δ -functor $(T_n, \delta_n)_{n \geq 0}$ (resp. $(T^n, \delta^n)_{n \geq 0}$) from $\mathcal{A}$ to $\mathcal{B}$ consists of

(i) a sequence of additive functors $(T_n : \mathcal{A} \rightarrow \mathcal{B})_{n \geq 0}$ resp. $(T^n : \mathcal{A} \rightarrow \mathcal{B})_{n \geq 0}$.

(ii) A sequence of natural transformations

$$\begin{array}{ccc} \text{Ex}_{\mathcal{A}} & \begin{array}{c} \xrightarrow{T_{n+1} \circ \text{pr}_3} \\ \parallel \delta_n \\ \xrightarrow{T_n \circ \text{pr}_1} \end{array} & \mathcal{B} \\ \text{resp.} & & \\ \text{Ex}_{\mathcal{A}} & \begin{array}{c} \xrightarrow{T^n \circ \text{pr}_3} \\ \parallel \delta^n \\ \xrightarrow{T^{n+1} \circ \text{pr}_1} \end{array} & \mathcal{B} \end{array}$$

such that this data assigns to each $\underline{\mathcal{E}} : 0 \rightarrow A_1 \xrightarrow{f} A_2 \xrightarrow{g} A_3 \rightarrow 0$ in $\text{Ex}_{\mathcal{A}}$ a long exact sequence in $\mathcal{B}$

$$\cdots \rightarrow T_{n+1} \xrightarrow{\delta_n^{\underline{\mathcal{E}}}} T_n A_1 \xrightarrow{T_n f} T_n A_2 \xrightarrow{T_n g} T_n A_3 \xrightarrow{\delta_{n-1}^{\underline{\mathcal{E}}}} T_{n-1} A_1 \rightarrow \cdots$$

resp.

$$\cdots \rightarrow T^{n-1} A_3 \xrightarrow{\delta_{\mathcal{E}}^{n-1}} T^n A_1 \xrightarrow{T^n f} T^n A_2 \xrightarrow{T^n g} T^n A_3 \xrightarrow{\delta_{\mathcal{E}}^n} T^{n+1} A_1 \rightarrow \cdots$$

Moreover the assignment $\mathcal{E} \rightarrow \text{l.e.s.}$ is functorial in $\text{Ex}_{\mathcal{A}}$

(iii) TODO WTF IS THIS?

Example 0.69. TODO

Construction 0.70. Suppose $\mathcal{A}$ has enough injectives and let $F : \mathcal{A} \rightarrow \mathcal{B}$ be a left exact additive functor.

1. $\forall A \in \mathcal{A}$ choose an injective resolution $\iota_A : \underline{A} \rightarrow I_A^\bullet$ in $\text{Ch}_{\geq 0}^*(\mathcal{A})$ and define $R^n F(A) := H^n(FI_A^\bullet)$ choice in $\mathcal{B}$ of n -th cohomology object.
2. $\forall f : A \rightarrow A'$ morphism in $\mathcal{A}$ choose an arrow ι_f such that

$$\begin{array}{ccc} \underline{A}' & \xrightarrow{\iota_{A'}} & I_{A'}^\bullet \\ f \uparrow & & \uparrow \iota_f \\ \underline{A} & \xrightarrow{\iota_A} & I_A^\bullet \end{array}$$

commutes in $\text{Ch}_{\geq 0}^*(\mathcal{A})$ which implies that $F(\iota_f) : FI_A^\bullet \rightarrow FI_{A'}^\bullet$ morphism in $\text{Ch}_{\geq 0}^*(\mathcal{B})$. Define: $R^n F(f) := H^n(F(\iota_f)) : R^n F(A) \rightarrow R^n F(A')$...

Lemma 0.71. (a) $R^n F$ is a functor $\mathcal{A} \rightarrow \mathcal{B}$ (additive.)

(b) If one makes other choices of injective resolutions $\tilde{\iota}_A : A \rightarrow \tilde{I}_A^\bullet$ and $\tilde{\iota}_f : \tilde{I}_A^\bullet \rightarrow \tilde{I}_{A'}^\bullet$, then we get a natural isomorphism $R^n F \cong \tilde{R}^n F$

3. Given $\mathcal{E} : 0 \rightarrow A_1 \xrightarrow{f} A_2 \xrightarrow{g} A_3 \rightarrow 0$ in $\text{Ex}_{\mathcal{A}}$ and an injective resolution $\iota_{\mathcal{E}} : \mathcal{E} \rightarrow J_{\mathcal{E}}^\bullet$ in $\text{Ch}_{\geq 0}^*(\text{Ex}_{\mathcal{A}})$

Lemma 0.72. (a) $FJ_{\mathcal{E}}^\bullet$ lies in $\text{Ch}_{\geq 0}^*(\text{Ex}_{\mathcal{B}})$, in particular, it is a s.e.s. of complexes in $\text{Ch}_{\geq 0}^*(\mathcal{B})$.

- (b) Write $J_{\mathcal{E}}^{\bullet} = 0 \rightarrow I_1^{\bullet} \rightarrow I_2^{\bullet} \rightarrow I_3^{\bullet} \rightarrow 0$ as a s.e.s. in $\text{Ch}_{\geq 0}^*(\mathcal{A})$, then the I_i^n are injective and by (a) we have a s.e.s.

$$0 \rightarrow FI_1^{\bullet} \rightarrow FI_2^{\bullet} \rightarrow FI_3^{\bullet} \rightarrow 0 \quad (*)$$

in $\text{Ch}_{\geq 0}^*(\mathcal{B})$, then the following diagram commutes and the top row is a l.e.s. in $\mathcal{B}$, and the vertical maps are isomorphisms.

$$\begin{array}{ccccccc} \cdots & \longrightarrow & H^{n-1}(FI_j^{\bullet}) & \xrightarrow{\delta_{FJ_{\mathcal{E}}}^{n-1}} & H^n(FI_1^{\bullet}) & \longrightarrow & H^n(FI_2^{\bullet}) \longrightarrow H^n(FI_3^{\bullet}) \xrightarrow{\delta_{FJ_{\mathcal{E}}}^n} \cdots \\ & & u_{A_3}^{n-1} \uparrow & & u_{A_1}^n \uparrow & & u_{A_2}^n \uparrow & & u_{A_3}^n \uparrow \\ \cdots & \longrightarrow & R^{n-1}F(A_3) & \longrightarrow & R^nF(A_1) & \xrightarrow{R^nF(f)} & R^nF(A_2) & \xrightarrow{R^nF(g)} & R^nF(A_3) \longrightarrow \cdots \end{array}$$

This is not even funny anymore. Define $\delta_{RF}^n : R^nF \circ \text{pr}_3 \Rightarrow R^nF \circ \text{pr}_1$ for $\mathcal{E}$ as $(u_{A_1}^{n+1})^{-1} \circ \delta_{FJ_{\mathcal{E}}}^n \circ u_{A_3}^n$.

4. Show that δ_{RF}^n is well defined.

Lemma 0.73. *TODO*

Theorem 0.74. *Suppose $\mathcal{A}$ has enough injectives and F is an additive left exact functor, then $RF := (R^nF, \delta_{RF}^n)_{n \geq 0}$ is a universal cohomological δ -functor and R^nF is called the n -th right derived functor of F . It satisfies:*

- (a) $R^0F = F$.
(b) $I \in \text{Inj}_{\mathcal{A}} \implies \forall n \geq 1, R^nF(I) = 0$.

Suppose $\mathcal{A}$ has enough projectives, let $G : \mathcal{A} \rightarrow \mathcal{B}$ be right exact, then

- (a) $\exists$ homological δ -functor $LG = (L_iG, \delta_i)_{i \geq 0}$ such that $(*)$

- (i) $L_0G = G$
(ii) $P \in \text{Proj}_{\mathcal{A}} \implies \forall n \geq 1, L_nG(P) = 0$.

- (b) LG with $(*)$ is universal.

- (c) LG with $(*)$ is unique up to unique isomorphism.

- (d) LG can be computed via projective resolutions, i.e. $\forall A \in \mathcal{A}$ with projective resolution $P^{\bullet} \rightarrow A$ in $\text{Ch}_{\leq 0}^*(\mathcal{A})$ we have

$$L_iG(A) = H^{-i}(GP^{\bullet})$$

Lemma. *If $T = (T_n, \delta_n)_{n \geq 0}$ is a homological δ -functor from $\mathcal{A}$ to $\mathcal{B}$ and if $\mathcal{A}$ has enough projectives and $T_iP = 0, \forall P \in \text{Proj}_{\mathcal{A}}, i \geq 1$, then T is universal.*

0.6 The Ext Functor

Let $\mathcal{A}$ be abelian, $M, N \in \mathcal{A}$ and suppose $\mathcal{A}$ has enough projectives/injectives if needed.

Definition 0.75. Define

$$\mathrm{Ext}_{\mathcal{A}}^i(-, N) := R^i \mathrm{Hom}_{\mathcal{A}}(-, N)$$

with the natural transformations δ^i .

Example. To compute: if

$$\cdots \rightarrow P^{-2} \rightarrow P^{-1} \rightarrow P^0 \rightarrow M \quad (*)$$

is a projective resolution, then we have the complex $P^\bullet$, apply $\mathrm{Hom}_{\mathcal{A}}(-, N)$ and get

$$0 \rightarrow \mathrm{Hom}_{\mathcal{A}}(P^0, N) \rightarrow \mathrm{Hom}_{\mathcal{A}}(P^{-1}, N) \rightarrow \mathrm{Hom}_{\mathcal{A}}(P^{-2}, N) \rightarrow \cdots$$

and $\mathrm{Ext}_{\mathcal{A}}^i(M, N) = H^i(\mathrm{Hom}_{\mathcal{A}}(P^\bullet, N))$

Proposition 0.76. (a) M projective $\implies \mathrm{Ext}_{\mathcal{A}}^i(M, N) = 0, \forall i \geq 1$.

(b) N injective $\implies \mathrm{Ext}_{\mathcal{A}}^i(M, N) = 0, \forall i \geq 1$.

Definition 0.77. Define

$$\overline{\mathrm{Ext}}_{\mathcal{A}}^i(M, -) := R^i(\mathrm{Hom}_{\mathcal{A}}(M, -))$$

with the natural transformations δ^i .

Example. Compute $\overline{\mathrm{Ext}}_{\mathcal{A}}^i(M, -)$ via injective resolutions of the 2nd argument:

$$0 \rightarrow N \rightarrow I^0 \rightarrow I^1 \rightarrow I^2 \rightarrow \cdots$$

and we get

$$\mathrm{Hom}_{\mathcal{A}}(M, I^0) \rightarrow \mathrm{Hom}_{\mathcal{A}}(M, I^1) \rightarrow \mathrm{Hom}_{\mathcal{A}}(M, I^2) \rightarrow \cdots$$

so $\overline{\mathrm{Ext}}_{\mathcal{A}}^i(M, -) = H^i(\mathcal{A}(M, I^\bullet))$.

Proposition 0.78. For M projective or N injective we have $\overline{\mathrm{Ext}}_{\mathcal{A}}^i(M, N) = 0, \forall i \geq 1$.

Remark. If $\mathcal{A}$ has enough projectives and injectives, $\mathrm{Ext}_{\mathcal{A}}^i(-, -)$ and $\overline{\mathrm{Ext}}_{\mathcal{A}}^i(-, -)$ turn out to be isomorphic!

Remark. $\mathrm{Ext}_{\mathcal{A}}^i(-, -)$ and $\overline{\mathrm{Ext}}_{\mathcal{A}}^i(-, -)$ are bifunctors.

Theorem 0.79. Suppose $\mathcal{A}$ has enough injectives and projectives, then $\exists$ natural isomorphisms as bifunctors

$$\begin{aligned} \mathcal{A}^{\mathrm{op}} \times \mathcal{A} &\rightarrow \mathbf{Ab} \\ u_{M,N}^i : \mathrm{Ext}_{\mathcal{A}}^i(M, N) &\rightarrow \overline{\mathrm{Ext}}_{\mathcal{A}}^i(M, N) \end{aligned}$$

0.6.1 Classical interpretation of Ext^1 as extension objects

Definition 0.80. (a) An extension of M by N is a s.e.s. in $\mathcal{A}$:

$$\mathcal{E} : 0 \rightarrow N \rightarrow E \rightarrow M \rightarrow 0$$

(b) Extensions $\mathcal{E}$ and

$$\mathcal{E}' : 0 \rightarrow N \rightarrow E' \rightarrow M \rightarrow 0$$

are called *equivalent* (write $\sim$), iff $\exists$ commutative diagram:

$$\begin{array}{ccccccccc} \mathcal{E} : 0 & \longrightarrow & N & \longrightarrow & E & \longrightarrow & M & \longrightarrow & 0 \\ & & \downarrow \text{id}_N & & \downarrow \varphi & & \downarrow \text{id}_M & & \\ \mathcal{E}' : 0 & \longrightarrow & N & \longrightarrow & E' & \longrightarrow & M & \longrightarrow & 0 \end{array}$$

(c) $\text{Ext}_{\mathcal{A}}^1(M, N)$ denotes the set of equivalence classes of extensions of M by N .

Remark. (a) By the snake lemma: $\mathcal{E} \sim \mathcal{E}' \implies E \cong E'$.

(b) $\sim$ is an equivalence relation.

Proposition 0.81. (a) $\text{Ext}_{\mathcal{A}}(M, N)$ is an abelian group for addition $\mathcal{E} + \mathcal{E}'$ defined by the Baer sum:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & N \oplus N & \longrightarrow & E \oplus E' & \longrightarrow & M \oplus M & \longrightarrow & 0 \\ & & \downarrow \Sigma & & \downarrow & & \downarrow \text{id}_{M \oplus M} & & \\ 0 & \longrightarrow & N & \longrightarrow & E \oplus E' \amalg_{N \oplus N} N & \longrightarrow & M \oplus M & \longrightarrow & 0 \\ & & \uparrow \text{id}_N & & \uparrow & & \uparrow \Delta & & \\ 0 & \longrightarrow & N & \longrightarrow & E + E' & \longrightarrow & M & \longrightarrow & 0 \end{array}$$

where $E \oplus E' \amalg_{N \oplus N} N$ is the pushout by the sum map $+: N \oplus N \rightarrow N, (n, m) \mapsto n + m$ and $E + E'$ is the pullback by the diagonal map $\Delta : M \rightarrow M \oplus M, m \mapsto (m, m)$. And the zero object is given by the split s.e.s

$$0 \rightarrow N \rightarrow N \oplus M \rightarrow M \rightarrow 0$$

(b) One has a natural isomorphism as bifunctors

$$u : \text{Ext}_{\mathcal{A}}(-, -) \Rightarrow \text{Ext}_{\mathcal{A}}^1(-, -)$$

if $\mathcal{A}$ has enough projectives (or to $\overline{\text{Ext}}_{\mathcal{A}}^1(-, -)$ if $\mathcal{A}$ has enough injectives).

Remark. In fact, Yoneda also considered higher Ext-groups (in the absence of injectives/projectives), e.g. $\text{Ext}^2(M, N)$ is the “group” of exact sequences

$$0 \rightarrow N \rightarrow E_1 \rightarrow E_2 \rightarrow M \rightarrow 0$$

modulo a suitable equivalence relation.

Notation. For ${}_R\text{Mod}$ one usually abbreviates

$$\text{Ext}_R^i(-, -) := \text{Ext}_{{}_R\text{Mod}}^i(-, -)$$

For $R = \mathbb{Z}[G]$ and a group G , one also considers the group cohomology for $M \in {}_{\mathbb{Z}[G]}\mathbf{Mod}$ as

Definition 0.82. Define

$$H^i(G, M) := \mathrm{Ext}_{\mathbb{Z}[G]}^i(\mathbb{Z}, M)$$

such that $\mathbb{Z}$ is the trivial G -action.

Proposition. In face $H^i(G, M)$ is the i -th right derived functor of

$$\begin{aligned} M \rightarrow \mathrm{Hom}_{\mathbb{Z}[G]}(\mathbb{Z}, M) &\stackrel{\text{exer.}}{=} M^G := \{m \in M \mid \forall g \in G : gm = m\} \\ \varphi_m : n \mapsto nm &\leftarrow m \in M^G \end{aligned}$$

So far we had $\mathrm{Ext}_{\mathcal{A}}^i$ valued in $\mathbf{Ab}$. But recall that $\mathrm{Hom}_R(-, -)$ is also a bifunctor

$$\begin{aligned} \mathrm{Hom}_R(-, -) : {}_R\mathbf{Mod}_{R'} \times {}_R\mathbf{Mod}_{R''} &\rightarrow {}_{R'}\mathbf{Mod}_{R''} \\ (M, N) &\mapsto \mathrm{Hom}_R(M, N) \end{aligned}$$

Fact: ${}_R\mathbf{Mod}_{R'}$ and ${}_R\mathbf{Mod}_{R''}$ have enough injectives and projectives because ${}_R\mathbf{Mod}_{R'}$ is isomorphic to ${}_{R \otimes (R')^{\mathrm{op}}}\mathbf{Mod}$.

Proposition 0.83 (Exer). *This Bifunctor induces*

$$\mathrm{Ext}_R^i(-, -) : {}_R\mathbf{Mod}_{R'} \times {}_R\mathbf{Mod}_{R''} \rightarrow {}_{R'}\mathbf{Mod}_{R''}$$

0.7 The Tor Functor

Let R be a ring, $M \in \mathbf{Mod}_R$ and $N \in {}_R\mathbf{Mod}$. We had the bifunctor

$$- \otimes - : \mathbf{Mod}_R \times {}_R\mathbf{Mod} \rightarrow \mathbf{Ab}$$

and it is right exact, and also $\mathbf{Mod}_R$ and ${}_R\mathbf{Mod}$ have enough projectives.

Definition 0.84. For $i \geq 0$ define:

$$\begin{aligned} \mathrm{Tor}_i^R(M, -) &:= L_i(M \otimes_R -) \\ \overline{\mathrm{Tor}}_i^R(-, N) &:= L_i(- \otimes_R N) \end{aligned}$$

Proposition 0.85. *If $P^\bullet \rightarrow M$ (or $Q^\bullet \rightarrow N$) are projective resolutions, then one has:*

$$\begin{aligned} \overline{\mathrm{Tor}}_i^R(M, N) &= H^{-i}(P^\bullet \otimes_R N) \\ \mathrm{Tor}_i^R(M, N) &= H^{-i}(M \otimes_R Q^\bullet) \end{aligned}$$

Moreover, if M or N are projective, then

$$\mathrm{Tor}_i^R(M, N) \cong 0 \cong \overline{\mathrm{Tor}}_i^R(M, N)$$

Theorem 0.86. *One has natural isomorphism of bifunctors*

$$\mathrm{Tor}_i^R(-, -) \cong \overline{\mathrm{Tor}}_i^R(-, -)$$

Other ways to compute Tor_i , not using projective resolutions.

Definition 0.87. Let $T = (T_n, \delta_n)$ be a homological δ -functor from $\mathcal{A}$ to $\mathcal{B}$. Call $A \in \mathcal{A}$ T -acyclic if $T_i A = 0, \forall i \geq 1$ (with T_0 right exact) For example M projective $\implies M$ is $L(- \otimes_R N)$ -acyclic

Facts 0.88. Let $0 \rightarrow A' \rightarrow A \rightarrow A'' \rightarrow 0$ be a s.e.s. in $\mathcal{A}$, then

(a) A'' is T -acyclic then

$$0 \rightarrow T_0 A' \rightarrow T_0 A \rightarrow T_0 A'' \rightarrow 0$$

is a s.e.s. (because $T_1 A'' = 0$ and l.e.s. from δ -functor).

(b) If A and A'' are T -acyclic, then so is A' (because of the l.e.s. from being a δ -functor).

$$\rightarrow T_2 A' \rightarrow \underbrace{T_2 A}_{=0} \rightarrow \underbrace{T_2 A''}_{=0} \rightarrow T_1 A' \rightarrow \underbrace{T_1 A}_{=0} \rightarrow \underbrace{T_1 A''}_{=0} \rightarrow T_0 A' \rightarrow T_0 A \rightarrow T_0 A''$$

(c) If $0 \rightarrow A_0 \rightarrow A_1 \rightarrow \dots \rightarrow A_n \rightarrow 0$ is exact in $\mathcal{A}$ and if $A_1, \dots, A_n$ are T -acyclic then A_0 is also T -acyclic. This follows from (b) by ind. using the exact sequences $0 \rightarrow A_0 \rightarrow A_1 \rightarrow X \rightarrow 0$ and

$$0 \rightarrow X \rightarrow A_2 \rightarrow \dots \rightarrow A_n \rightarrow 0$$

Lemma 0.89. If $C^\bullet \in \text{Ch}_{\leq 0}^*(\mathcal{A})$ is acyclic with all C^i T -acyclic, then $T_0 C^\bullet$ is acyclic (we assume T_0 is right exact)

Corollary 0.90. Suppose $\mathcal{A}$ has enough projectives, $G : \mathcal{A} \rightarrow \mathcal{B}$ is right exact and $T = LG$. If $Q^\bullet \xrightarrow{g} \underline{A}$ is a resolution by T -acyclic objects. Then:

$$L_i G(A) \cong H^{-i}(GQ^\bullet)$$

Definition 0.91. $M \in \text{Mod}_R$ is called *flat* if $M \otimes_R - : {}_R\text{Mod} \rightarrow \text{Ab}$ is exact.

Proposition 0.92. For $M \in \text{Mod}_R$ the following are equivalent:

- (a) M is flat.
- (b) $\text{Tor}_1^R(M, -) = 0$.
- (c) $\text{Tor}_i^R(M, -) = 0 \forall i \geq 1$.

Theorem 0.93. For $M \in \text{Mod}_R$ the following are equivalent:

- (a) M is flat.
- (b) $\forall N \in {}_R\text{Mod} : M$ is $L(- \otimes_R N)$ -acyclic.

And thus Tor_i^R can be computed via flat resolutions.